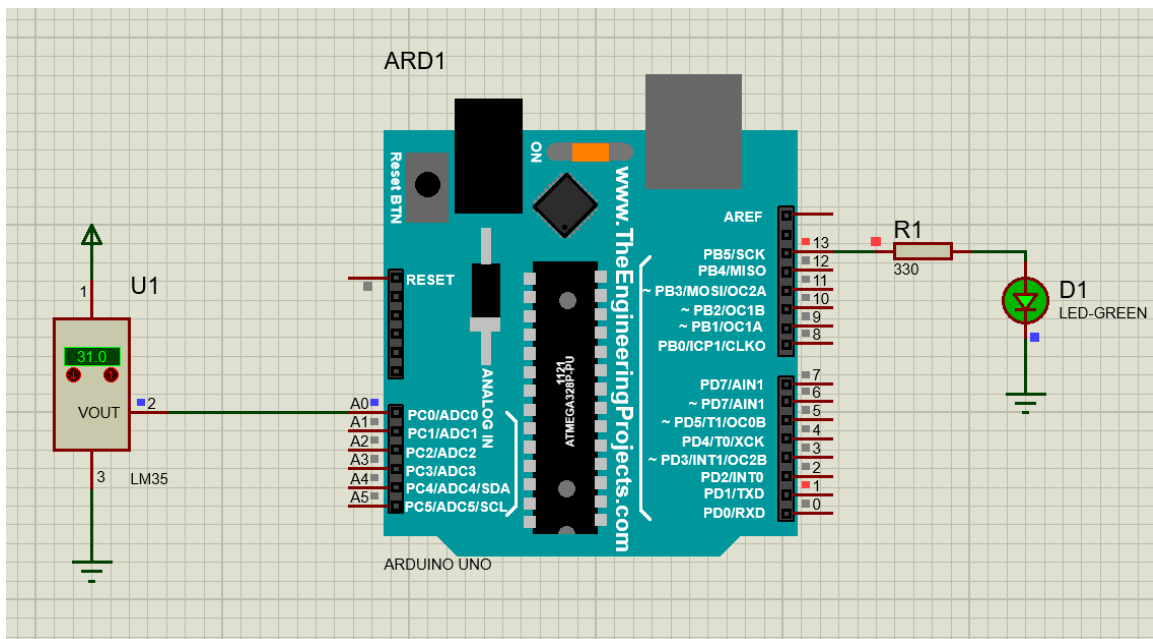
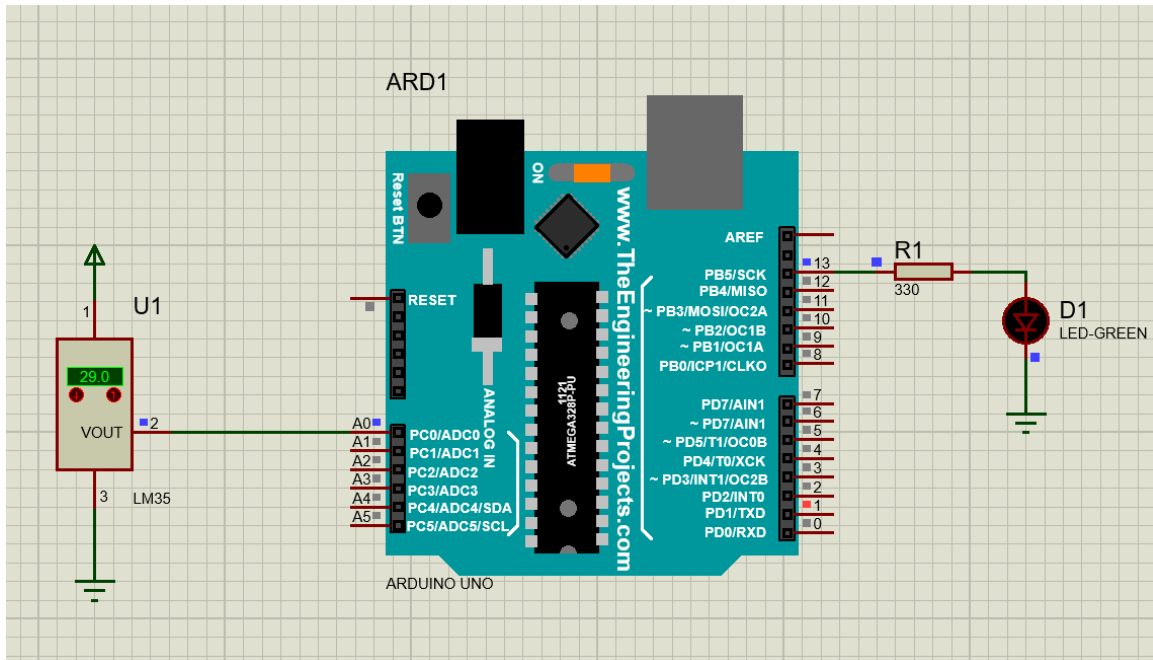


The Simulation in proteus:



The Code:

```
int sensorPin = A0;
int ledPin = 13;

void setup()
{
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int sensorValue = analogRead(sensorPin);

  float voltage = sensorValue * (5.0 / 1023.0);

  float temperature = voltage * 100.0;

  Serial.print("Temperature = ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature > 30)
  {
    digitalWrite(ledPin, HIGH);
  }
  else
  {
    digitalWrite(ledPin, LOW);
  }

  delay(500);
}
```

Code explanation:

The Arduino program reads temperature values from the **LM35 temperature sensor** and controls an LED based on the measured temperature.

1. Variable Declaration

```
int sensorPin = A0;
int ledPin = 13;
```

- `sensorPin` stores the analog pin connected to the LM35 sensor.
 - `ledPin` stores the digital pin connected to the LED.
-

2. Setup Function

```
void setup()
{
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}
```

- `pinMode(ledPin, OUTPUT);` configures pin 13 as an output to control the LED.
 - `Serial.begin(9600);` starts serial communication at 9600 baud rate, allowing temperature values to be displayed on the Serial Monitor.
-

3. Reading Sensor Data

```
int sensorValue = analogRead(sensorPin);
```

- The Arduino reads the analog voltage produced by the LM35 sensor.
 - The value returned ranges from **0 to 1023**, corresponding to **0V to 5V**.
-

4. Converting ADC Value to Voltage

```
float voltage = sensorValue * (5.0 / 1023.0);
```

- The analog reading is converted into a voltage value.
 - Since the Arduino uses a 10-bit ADC, the maximum reading is 1023.
 - This formula calculates the actual sensor output voltage.
-

5. Calculating Temperature

```
float temperature = voltage * 100.0;
```

- The LM35 outputs **10 mV (0.01 V) per degree Celsius**.
- Multiplying the voltage by 100 converts it into temperature in Celsius.

Examples:

- 0.25 V → 25°C
 - 0.30 V → 30°C
 - 0.35 V → 35°C
-

6. Displaying Temperature

```
Serial.print("Temperature = ");  
Serial.print(temperature);  
Serial.println(" C");
```

- The temperature value is displayed on the Serial Monitor.
 - This helps users monitor temperature readings in real time.
-

7. LED Control Logic

```
if (temperature > 30)  
{  
    digitalWrite(ledPin, HIGH);  
}  
else  
{  
    digitalWrite(ledPin, LOW);  
}
```

- If the temperature exceeds **30°C**, the LED turns ON.
 - If the temperature is **30°C or below**, the LED turns OFF.
 - The LED acts as a warning indicator for high temperature.
-

8. Delay

```
delay(500);
```

- The Arduino waits for 500 milliseconds before taking the next reading.
 - This prevents excessive sensor readings and provides stable monitoring.
-

System Operation Summary

1. The LM35 sensor measures temperature.
2. Arduino reads the sensor value through pin A0.
3. The analog value is converted into voltage.

4. Voltage is converted into temperature ($^{\circ}\text{C}$).
5. The temperature is displayed on the Serial Monitor.
6. If the temperature exceeds 30°C , the LED turns ON; otherwise, it remains OFF.
7. The process repeats every 500 ms.

This simulation demonstrates a simple **IoT-based healthcare monitoring system**, where abnormal temperature levels are detected and indicated automatically using an LED alert.